

HDOC – Experiment 37

Question

Write a menu-driven C program to perform the following operations using separate functions:

1. Convert a lowercase string to uppercase without using built-in functions.
2. Reverse a string.
3. Check if a string is a palindrome.
4. Exit.

❖ Algorithm

- 1 Start the program.
- 2 Display the menu containing the four choices.
- 3 Read the user's choice.
- 4 Use switch to perform the selected operation:
 - **Choice 1:** Read a string. Traverse each character and, if it lies between 'a' and 'z', convert it to uppercase using character arithmetic.
 - **Choice 2:** Read a string, calculate its length manually, and print its characters from the last character to the first.
 - **Choice 3:** Read a string and calculate its length manually. Compare the first character with the last, second with second-last, and so on. If every pair matches, the string is a palindrome.
 - **Choice 4:** Exit the program.
- 5 For an invalid choice, display an error message.
- 6 Repeat the menu until the user selects **4**.
- 7 Stop.

❖ Test case table

Test	Choice	Input	Expected Output
1	1	hello world	HELLO WORLD
2	1	computer	COMPUTER
3	2	hello	olleh
4	2	OpenAI	IAnepO
5	3	madam	Palindrome
6	3	level	Palindrome
7	3	hello	Not a palindrome
8	5	—	Invalid choice
9	4	—	Program exits

❖ Program code in C

```
#include <stdio.h>
```

```
void toUppercase();
```

```
void reverseString();
```

```
void checkPalindrome();
```

```
int main()
```

```
{
```

```
    int choice;
```

```
    do
```

```
{  
    printf("\n===== STRING OPERATIONS =====\n");  
    printf("1. Convert lowercase string to uppercase\n");  
    printf("2. Reverse a string\n");  
    printf("3. Check if a string is a palindrome\n");  
    printf("4. Exit\n");  
    printf("=====\n");  
    printf("Enter your choice: ");  
    scanf("%d", &choice);  
  
    switch (choice)  
    {  
        case 1:  
            toUppercase();  
            break;  
  
        case 2:  
            reverseString();  
            break;  
  
        case 3:  
            checkPalindrome();  
            break;  
  
        case 4:  
            printf("Exiting program...\n");  
            break;
```

```

        default:

            printf("Invalid choice! Please enter 1 to 4.\n");

        }

    } while (choice != 4);

    return 0;
}

/* Function to convert lowercase string to uppercase */
void toUppercase()
{
    char str[100];
    int i;

    printf("Enter a lowercase string: ");
    scanf("%s", str);

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - ('a' - 'A');
        }
    }

    printf("Uppercase string: %s\n", str);
}

```

```
/* Function to reverse a string */
```

```
void reverseString()
```

```
{
```

```
    char str[100];
```

```
    int i, length = 0;
```

```
    printf("Enter a string: ");
```

```
    scanf("%[^\n]", str);
```

```
    /* Find length without strlen() */
```

```
    while (str[length] != '\0')
```

```
    {
```

```
        length++;
```

```
    }
```

```
    printf("Reversed string: ");
```

```
    for (i = length - 1; i >= 0; i--)
```

```
    {
```

```
        printf("%c", str[i]);
```

```
    }
```

```
    printf("\n");
```

```
}
```

```
/* Function to check whether a string is a palindrome */
```

```
void checkPalindrome()
```

```
{  
    char str[100];  
    int i, length = 0;  
    int palindrome = 1;  
  
    printf("Enter a string: ");  
    scanf("%[^\n]", str);  
  
    /* Find length without strlen() */  
    while (str[length] != '\0')  
    {  
        length++;  
    }  
  
    for (i = 0; i < length / 2; i++)  
    {  
        if (str[i] != str[length - 1 - i])  
        {  
            palindrome = 0;  
            break;  
        }  
    }  
  
    if (palindrome)  
        printf("%s is a palindrome.\n", str);  
    else  
        printf("%s is not a palindrome.\n", str);  
}
```

❖ Compile, Execute and Evaluate

```
(kali㉿kali)-[~/Desktop/exp37]
$ vi string2.c

(kali㉿kali)-[~/Desktop/exp37]
$ gcc string2.c -o string2

(kali㉿kali)-[~/Desktop/exp37]
$ ./string2

===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit

Enter your choice: 1
Enter a lowercase string: hello world
Uppercase string: HELLO WORLD

===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit

Enter your choice: 1
Enter a lowercase string: computer
Uppercase string: COMPUTER

===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit

Enter your choice: 2
Enter a string: hello
Reversed string: olleh
```

```
===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
```

```
=====
Enter your choice: 2
Enter a string: OpenAI
Reversed string: IAnepO
```

```
===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
```

```
=====
Enter your choice: 3
Enter a string: madam
"madam" is a palindrome.
```

```
===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
```

```
=====
Enter your choice: 3
Enter a string: level
"level" is a palindrome.
```

```
===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
```

```
=====
Enter your choice: 3
Enter a string: hello
"hello" is not a palindrome.
```

```
===== STRING OPERATIONS =====  
1. Convert lowercase string to uppercase  
2. Reverse a string  
3. Check if a string is a palindrome  
4. Exit
```

```
=====
```

Enter your choice: 5
Invalid choice! Please enter 1 to 4.

```
===== STRING OPERATIONS =====  
1. Convert lowercase string to uppercase  
2. Reverse a string  
3. Check if a string is a palindrome  
4. Exit
```

```
=====
```

Enter your choice: 4
Exiting program...

```
(kali@kali)-[~/Desktop/exp37]  
$
```